

Final Audit Report

2026-05-14

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Final Audit Report

Run id: run-2026-05-14T030813Z

Status distribution

Plan milestones only:

Status	Count
validated	10
not-started	2
in-progress	0
action_required	0
deferred	0
reopened	0
superseded	0
invalidated	0

Non-plan items checked separately: _run/final-delivery-ready is validated / high; _manager/validator-warnings remains in-progress / medium and is not a terminal scientific claim.

Plan adherence

Milestone	Terminal status	Confidence	Evidence count	Final audit judgment
M-1	validated	high	8	Supported by candidate triage, proof-sketch, scaling data, figures, and tests.
M-2	not-started	provisional	0	No direct ledger event or standalone continuous norm-mismatch theorem. Later package files disclose this as residual debt.
M-3	validated	high	8	Supported by the collocation blind-spot theorem, certificate scripts, data, figures, and tests.
M-4	not-started	provisional	0	No direct ledger event or standalone conservation-law/shock-selection campaign. Later package files disclose partial absorption into admissibility analysis.

Milestone	Terminal status	Confidence	Evidence count	Final audit judgment
M-5	validated	high	10	Supported by reproducibility and figure manifests plus final-check scripts for the early collocation package.
M-6	validated	high	6	Scientific synthesis is supported by later package artifacts, but one historical ledger artifact pointer is absent. See FA-V1-001.
M-7	validated	high	3	Supported by the broad catalogue, application map, and toy-simulation plan.
M-8	validated	high	11	Supported by weak-topology branch artifacts, scripts, data, figures, and tests.
M-9	validated	high	7	Supported by admissibility/invariant branch artifacts and positivity/mass toy evidence.

Milestone	Terminal status	Confidence	Evidence count	Final audit judgment
M-10	validated	high	15	Supported by ODE reliability branch artifacts and hidden-mode, Lyapunov, and inverse-identifiability evidence.
M-11	validated	high	21	Supported by the toy-simulation result note and at least six focused toy simulations with data, figures, and tests.
M-12	validated	high	7	Supported by the broad synthesis package, catalogue, application map, toy results, and branch reports.

The run adhered to the broadened catalogue-oriented plan for M-7 through M-12. It did not complete standalone M-2 or M-4 as originally written, but the terminal package does not silently claim they were validated.

Confidence calibration

Validated plan milestones:

Confidence	Count
high	10
medium	0
low	0
provisional	0

Low-confidence terminal states: none among validated plan milestones.

Provisional terminal states: M-2 and M-4 are not-started / provisional, not validated. This is an honest residual-debt state rather than an overconfident validation.

Validator calibration: `promise_check` exited 0 with warnings, and `org_check` exited 0 with warnings. The warnings are nonblocking except for the missing M-6 ledger artifact pointer captured below.

Residual debt

Item	Kind	Narrative
M-2	not-started	No standalone continuous norm-mismatch counterexample was validated. Weak-topology and trace-leakage work partly cover adjacent mechanisms, but not the original M-2 theorem as stated.
M-4	not-started	No standalone conservation-law or shock-selection simulation/proof campaign was completed. Burgers/admissibility analysis partially covers the area, but the original milestone remains unvalidated.
M-6	missing_evidence_artifact	The latest validated M-6 event references <code>reports/final/final_report.md</code> , which is absent. Later artifacts preserve the scientific synthesis claim.
CAT-18	deferred	The inverse PDE/source-sensor variant remains deferred as a lower-value or harder branch.
CAT-20	deferred	The long-horizon rollout surrogate variant remains deferred and would require a separate finite-dimensional rollout/stability treatment.
<code>_manager/validator-warnings</code>	in-progress	Validator warnings persist as organizational/history warnings; they do not change the scientific milestone distribution.

Findings by severity

CRITICAL

None.

MODERATE

1. FA-V1-001 — M-6 — missing evidence artifact. The latest validated M-6 ledger event references `reports/final/final_report.md`, but that file is absent on disk. Later package artifacts, including `reports/residual_minimization_reliability_final_report.md` and `residual-certificates/broad_synthesis_package.md`, preserve the scientific claim and disclose old non-canonical/missing artifact warnings. This is evidence-traceability debt rather than scientific invalidation.

MINOR

None logged as structured final-audit findings. Validator and organization warnings are noted as residual debt where relevant.

Future work

Anchored to	Proposal
M-2	Either prove a standalone continuous noncoercive residual theorem with a precise corrected norm/certificate, or explicitly retire the original M-2 label in a future ledger correction.
M-4	Complete a scalar conservation-law entropy/shock-selection proof or toy campaign, or explicitly retire the standalone M-4 milestone label.
M-6	Emit a ledger correction or archive event replacing the missing <code>reports/final/final_report.md</code> pointer with the actual final report and package artifacts.
CAT-18	Decide whether inverse PDE/source-sensor nullspace failure warrants a standalone proof or toy demonstration.
CAT-20	If long-horizon rollout reliability matters, build a finite-dimensional rollout toy with a multi-step stability or invariant certificate.

Reconciliation log

No `_plan/`, `_archive/`, or correction events are emitted by this final-audit document stage.

Recommended reconciliation, not emitted here: repair the M-6 artifact pointer from `reports/final/final_report.md` to the actual final report path or add an explicit archive/correction ledger event.